



Revolutionizing Two-Wheelers

ADAS, Connectivity, Electrification

March 14 2024

Author:

Mrinal Sarkar

Senior Architect - System Software, Sasken Technologies Ltd.



SASKEN

Table of Content

1. Introduction

2. Exploring the Problem Statement

3. Solution

4. ARAS: For Motorcycle Rider Safety and Comfort

5. Connectivity

6. Electrification of the Motorcycle

7. Market Overview

8. Customers/Success Stories

9. Conclusion

10. About the Author

11. About Sasken Technologies

Abstract

This whitepaper delineates into the pivotal role of the two-wheeler industry within the broader automotive sector and explores emerging technologies aimed at enhancing safety, rider comfort, affordability, and environmental sustainability. It delves into current market dynamics, including market share analysis and key drivers propelling future growth. Furthermore, it elucidates how providers and service industries can align themselves with these technological advancements to capitalize on evolving consumer demands and market trends.

Introduction

Undoubtedly, the automotive industry is rapidly progressing towards speed, safety, and cost-effectiveness on a global scale. The automotive industry is fervently embracing cutting-edge technologies aimed at augmenting safety measures and promoting financial efficiency within our vehicular experiences. The need for swiftness arises from the popularity of 2-wheelers (also known as motorcycles) as the preferred medium of road transportation in both developing and developed countries. In countries like India, the scope of the 2-wheeler market is expansive considering that they are low-cost, fuel-efficient, and easy to steer through the traffic. So, it's no surprise that motorcycles are starting to take a front seat in driving the automotive market's growth.

Exploring the Problem Statement

One of the most critical challenges of riding motorcycles has been the vulnerability to road accidents. The build and the design of a motorcycle make it much lighter than cars and much faster than most vehicles in traffic conditions because of which they are more prone to accidents.

Apart from road accidents, motorcyclists are at risk due to reckless, careless, or distracted driving which includes - not following speed limits, not driving as per road/weather conditions, failing to pay proper attention to the road conditions, failing to wear safety gear, and being inexperienced.



Solution

There are a lot of technological measures that have been taken for cars and heavy vehicles to reduce the vehicle running cost and safety in transportation systems such as Advanced Driver Assistance Systems (ADAS), and Connectivity and Electric Vehicle (EV).

But a question comes to mind: whether the same can be effectively applied to the motorcycle as well or not?

The answer is, yes.

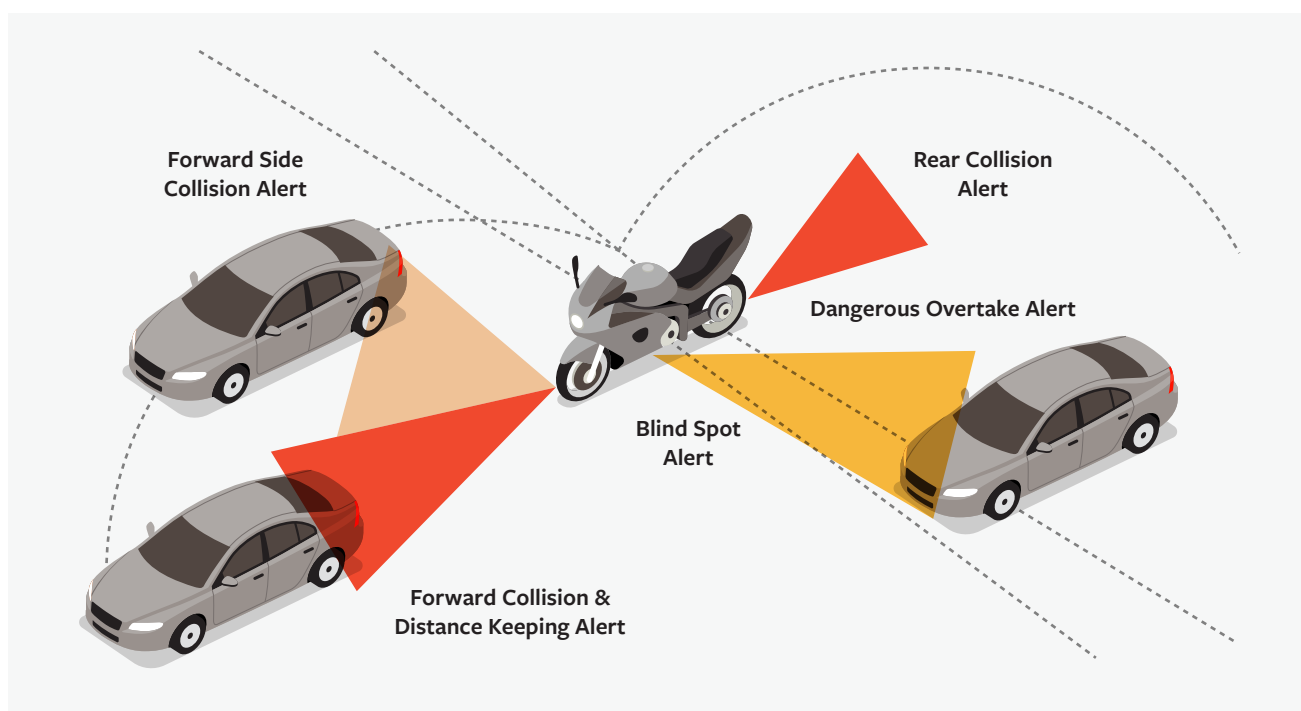
However, due to Advanced Rider Assistance Systems (ARAS) and Connectivity the purchase cost of the motorcycle would likely go up. On the other hand, it provides rider comfort, rider safety, and assistance to both rider and nearby vehicle drivers. Electrification of motorcycles will also further bring down running costs and carbon footprint.

ARAS: For Motorcycle Rider Safety and Comfort

The Motorcycle Advanced Riding Assistance System (ARAS) is based on the two most important technologies available today – Computer Vision by Machine Learning and Vehicle-to-Everything (V2X) communication.

ARAS aims to enhance motorcycle safety by using advanced algorithms that combine computer vision by machine learning and V2X wireless communication with nearby vehicles or infrastructure (like traffic lights) with real-time information. This thereby detects safety risks and imminent crash/danger to the rider.

ARAS analyzes real-time camera feeds from the front and rear cameras along with inputs from different sensors mounted on the motorcycle to create a computer vision.

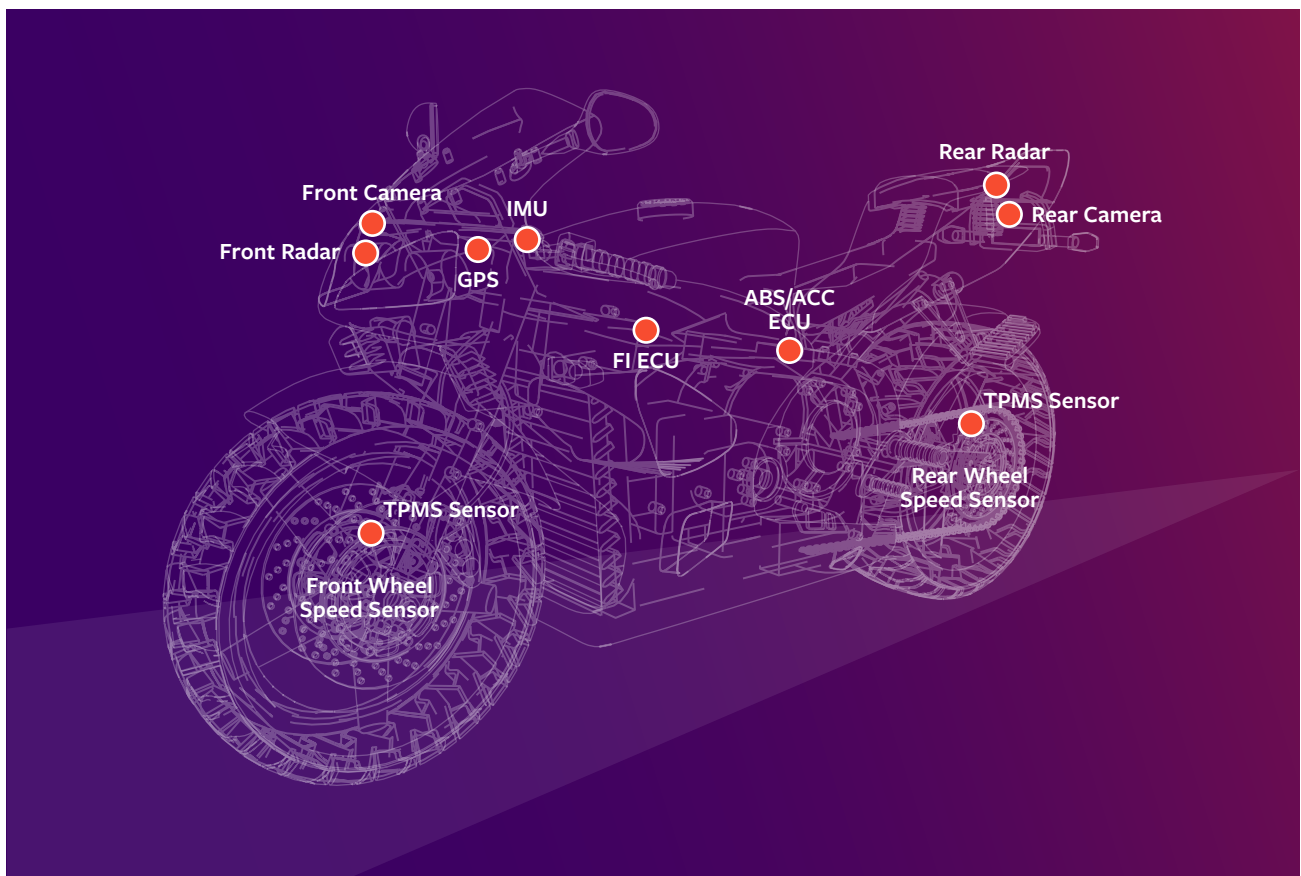


SENSORS THAT ARE USED IN ARAS

Camera sensors play a pivotal role in ARAS. It identifies and distinguishes objects, vehicles, people, and animals on the road and enables features like adaptive cruise control and lane departure warning.

Radar/LiDAR/Ultrasonic/Sonar sensors use radio waves and lasers to detect even smaller objects and highly precise distances from the objects. This information is crucial to make a judgment for adaptive cruise control, blind-spot monitoring, helping with parking maneuvers, and pedestrian detection.

GPS/GNSS sensors also contribute to precise positioning and orientation thus enhancing the overall situational awareness for ARAS.



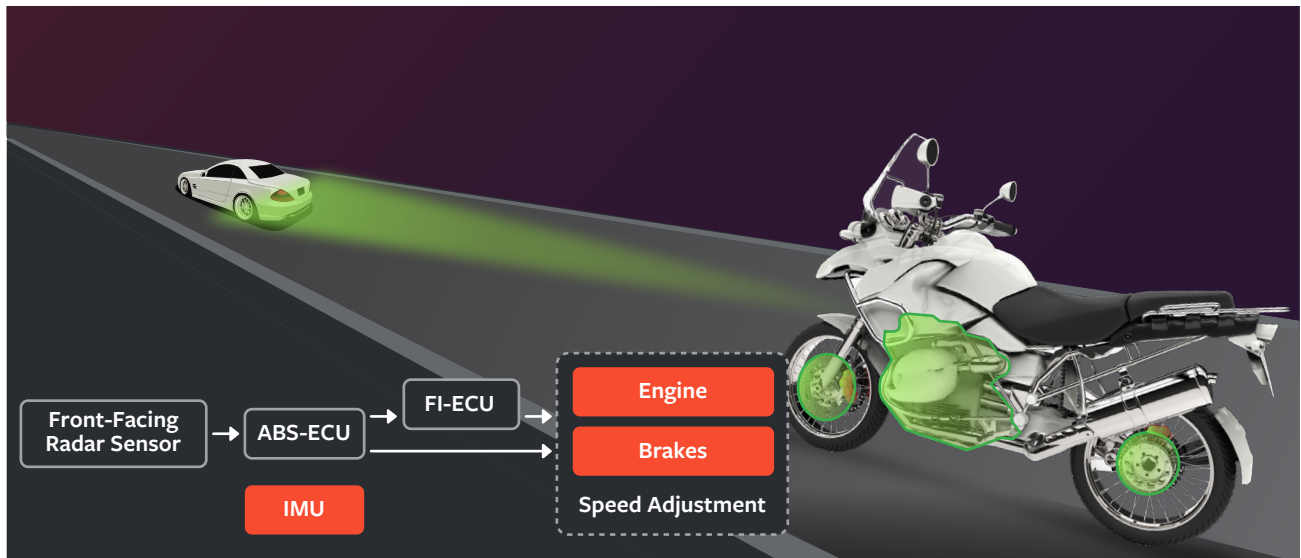
ARAS SAFETY FUNCTIONS

Front and Back Collision Avoidance Warning: The motorcyclist is warned with an audio-visual signal. The embedded ARAS system warns the rider of the potential risks of collision in case the proximity is closer (to vehicles, people, and animals) and initiates action.

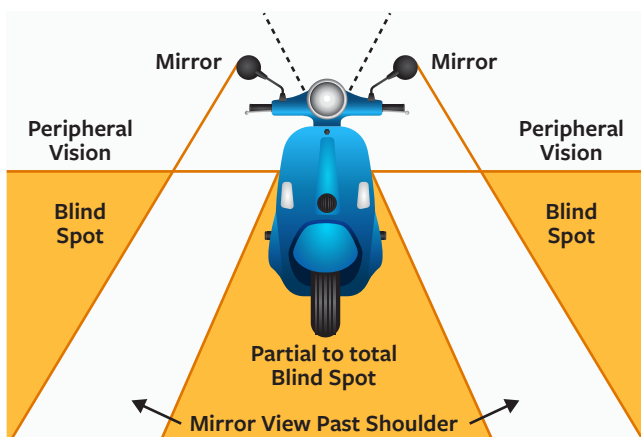
Lane Departure Alert: The lane departure alert system warns the motorist that the motorcycle is about to cross lane divider lines. Lane marking is of two colors - white and yellow and path markings can be either broken, strong, twofold strong, or a mix of the strong and broken path. The AI system in ARAS needs to identify and determine whether the rider is at risk of crossing any of these lines. Therefore the ARAS system on the motorcycle needs to be well trained with all different lane divider-related information.

Lane Keeping Assist System (LKAS): This system helps motorcycles return to their lane if motorists drift out without indicators. This could help to prevent a future collision.

Adaptive Cruise Control (ACC): A radar sensor on the front/back of the motorcycle determines the distance to preceding/succeeding vehicles using yaw rate and vehicle speed. If this distance reduces, then the system either automatically reduces the speed to reinstate the desired distance to preceding vehicles to avoid front collision or alerts the succeeding vehicle with flashing indicators to avoid back collision.



Blind Spot Detection: Blind spots are areas at a certain angle surrounding a motorcycle that the rider cannot see when looking into the side-view mirrors. Thus, often motorist turns their heads and check for any adjacent vehicles before they decide to change direction or lane. This leads to lots of front/side collision accidents in case riders turn or do not turn their heads to check the adjacent vehicles in blind spots. Blind-spot monitoring helps motorists be more aware of objects in these blind spots thus increasing safety by alerting the motorist about nearby objects/vehicles and not running into another vehicle moving in the same direction in an adjacent lane.



Automatic Headlight High-beam Activation and Dimming: This technology helps to solve two issues related to headlight high-beam activation and dimming otherwise the rider must control it manually.

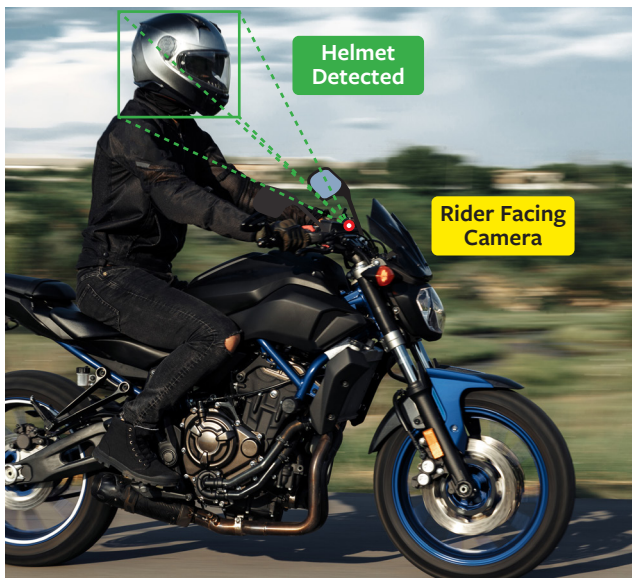
- Turns off the high beams lights to avoid blinding occupants of oncoming vehicles.
- Turns on the high beams lights when the road ahead is dark. This helps the motorist who doesn't always think to switch them on.

Motorcycle Autonomous Emergency Braking System (MAEB): This is a technology that can apply autonomously a braking force to reduce impact speed in emergencies based on the front camera feed and different speed sensors.

Anti-Lock Braking System (ABS): ABS uses speed sensors on both wheels to determine wheel speed and when a wheel is about to lock. In sudden braking situations, ABS adjusts the braking pressure accordingly to prevent the wheel from locking and maintain the stability of the motorcycle.

Tire Pressure Monitoring System (TPMS): Deflated tires cause most accidents on the road. TPMS can play a significant role in the prevention of future motorcycle accidents by providing real-time alerts on the pressure levels of your tires, improving handling and braking capabilities, and improving tire life and performance.

Rider's Safety Gears Detection: The most common cause of any motorcycle accident-led injuries/fatalities is due to not wearing safety gear (particularly a helmet) by the rider. Front rider-facing camera feed can help to detect whether the rider is wearing a helmet or not and accordingly alert the rider with audio and visual signals.



Reckless Driving Pattern Detection: Using different speed sensors, gyro sensors data, and camera feed, the ARAS system can detect reckless driving patterns and thus can alert the rider and nearby vehicles over V2V communication to avoid any collision.

Connectivity

With the help of connectivity via the Telematics unit in motorcycle, some of the important functions can be enabled which will further help in motorcycle rider safety and comfort.

Emergency call (eCall) – For after-crash support to the rider.

Remote Services – This helps motorists, dealers, OEMs, and the insurance company provide maintenance and insurance support. Remote services include theft alerts, stolen vehicle locator, real-time vehicle data upload to OEM servers for better vehicle and battery management (for EV), remote diagnostics, etc.

OTA Software update – Helps to update the software for different ECUs in motorcycles remotely without the owner visiting the dealer shop.

Map Data update – Helps in updating the navigation map data and provides updated POI information.

Electrification of the Motorcycle



The two-wheeler market consists of scooters and motorcycles. However, motorcycle holds the major portion of the market share. As of today, the electric scooter from different OEMs is the main point of focus, but the same electrification for motorcycles remains an untapped opportunity. Electrification of the motorcycle will majorly impact the 2-wheeler market as the operating cost and maintenance costs are going to be much lesser compared to conventional Internal Combustion Engine (ICE)-based motorcycles. Electric powertrains have the potential and a whole range of opportunities over ICE-based motorcycles to make the machine significantly more advanced due to software-driven architecture and connectivity. ICE vehicles require complex mechanical parts whereas, electric powertrains use electric motors, which require specialized electrical components, such as batteries, power electronics, and electric motors.

An electric motor has better performance in terms of Torque and Thermal management than an ICE-based engine -

- **Torque:** the electric motorcycle provides instant and consistent torque due to the inherent behavior of electric motors, but ICE-based motors require time to reach the peak torque and find it difficult to maintain the torque due to their mechanical parts. Thus, an electric motor gives a better riding experience.
- **Thermal management:** Electric engines can have a better thermal management system to maintain the optimal operating temperature of the engines than ICE-based engines. This, in turn, helps in better performance and durability.

Software-defined Architecture in E-motorcycles helps in better motor control, battery management, and connectivity which in turn provides real-time data on battery status, range, etc. Thus, making it easier for predictive maintenance and identifying potential issues that can lead to critical failures.

Enhanced Regenerative Braking (ERB) in E-motorcycles recaptures the kinetic energy being wasted during braking and further recharges the battery using that energy.

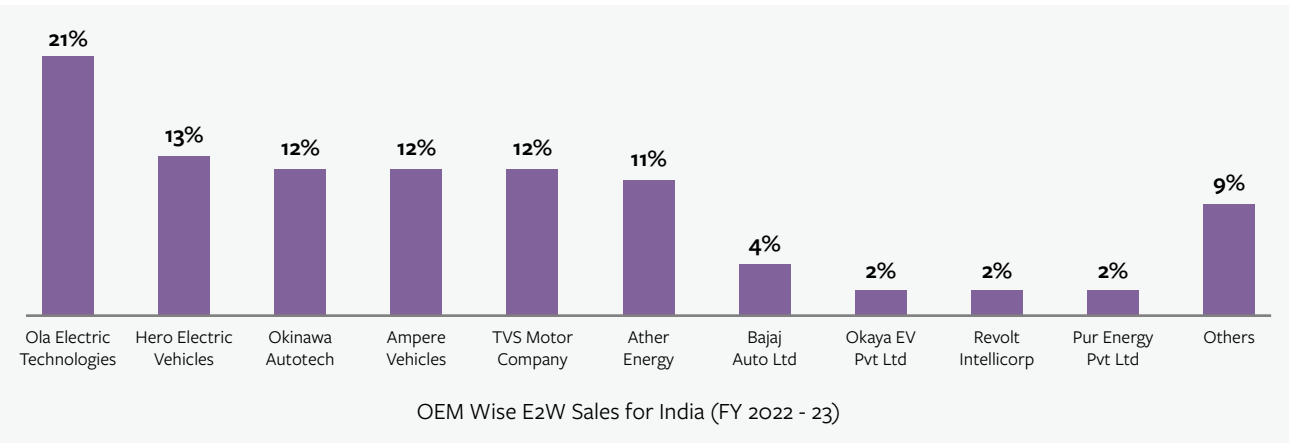
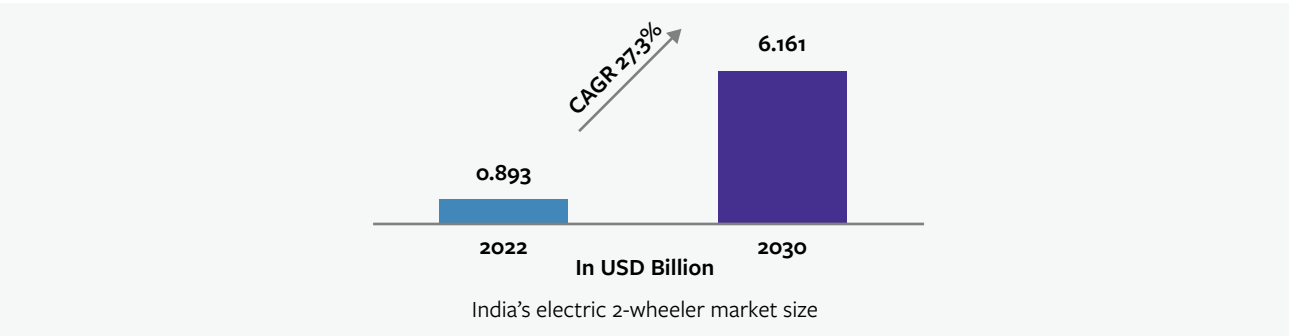
- **Motor Control:** Electric motorcycles will require enhanced motor controllers to control/regulate the power delivery to the electric motor. Advanced real-time edge computing software delivers enhanced performance and safety based on rider input and road conditions. It utilizes crucial data from RPM sensors, and torque output sensors to perform instant adjustments to motor response for power delivery optimization and efficiency.
- **Battery Management:** Electric motorcycles utilize an advanced Battery Management System (BMS) to monitor the "State of Health (SOH)" of the battery pack (a collection of battery cells). This system balances the cells to ensure uniform voltage levels are maintained across all cells, optimizes the use of residual energy present in the battery, and manages to deliver operational voltage and current consistently. Based on the charging status inputs from the battery management system, nearby charging stations can be indicated on the navigation unit.

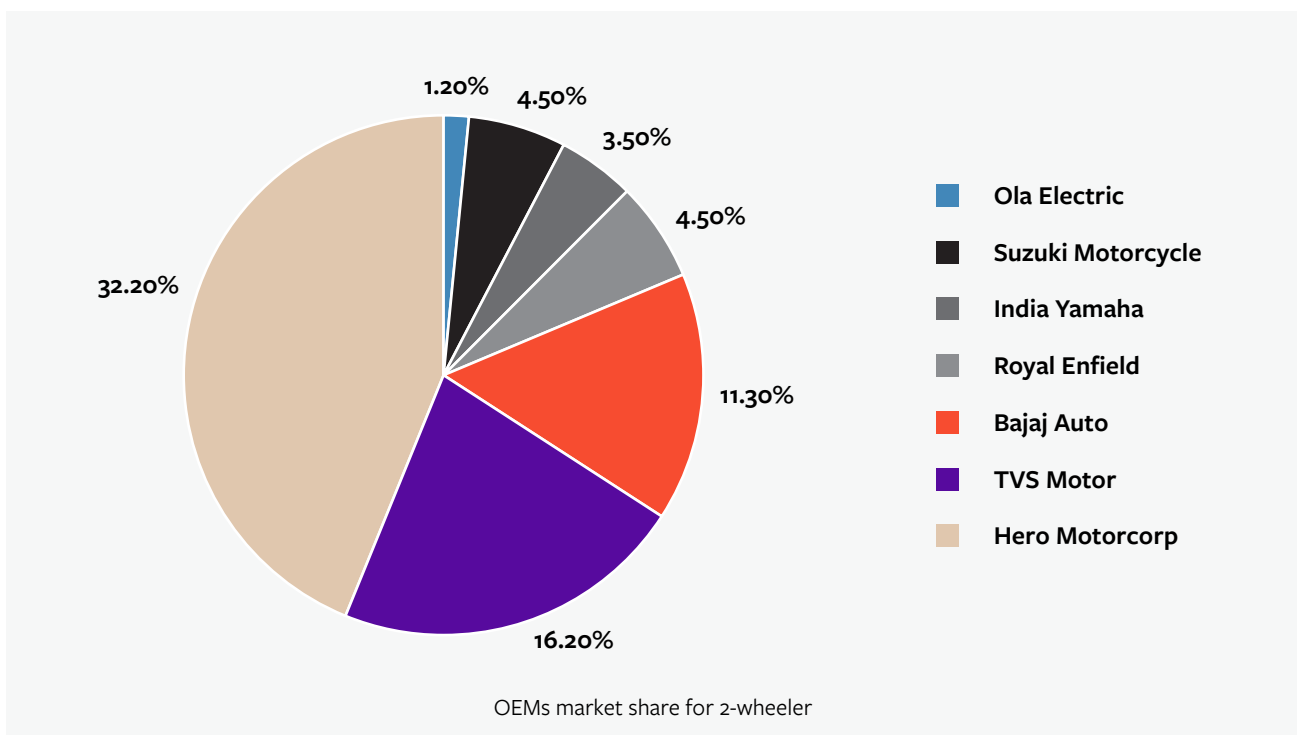
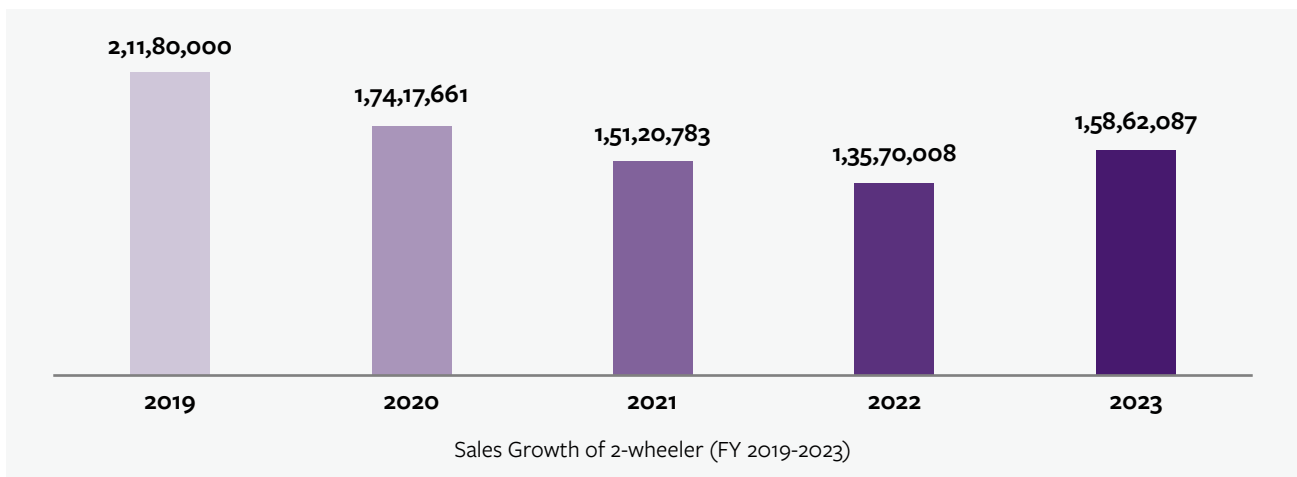
Market Overview:

Market: Indian electric two-wheeler OEMs are witnessing a transformative landscape marked by cutting-edge technologies like zero-emission, high-capacity Lithium-ion batteries, optimizing performance and charging infrastructure.

Investments: Key players invest substantially in R&D, ensuring affordability while attracting customers and driving sales which aligns with India's commitment to decarbonize transport, lower emissions, and achieve net zero by 2070.

Solutions: Indian electric two-wheeler OEMs are at the forefront of innovation, utilizing advanced battery technology, battery advancements like cobalt-free and silicon anode batteries, and smart connectivity to enhance vehicle performance.





KEY INITIATIVES:

Electric two-wheeler OEMs in India showcase a commitment to innovation through initiatives like design enhancements, technological upgrades, and extended-range offerings.

OEMs are making substantial investments in their product lines and in charging infrastructure. This holistic approach witnessed through partnerships and investments in charging networks, signifies a broader dedication to fostering the overall growth of the electric vehicle ecosystem.

The government has introduced a battery-swapping policy, coupled with interoperability standards, which aims to enhance the efficiency of battery-swapping ecosystems to reduce charging time, crucial factors in accelerating the adoption of electric two-wheelers.

KEY OPPORTUNITIES:

Indian electric two-wheeler startups are prioritizing sustainability by integrating electric vehicles into broader ecosystems with a focus on building a robust charging network, commercial EV financing, and lifecycle management platform.

Service Providers can adopt a holistic approach, actively engaging with government initiatives, strengthening the supply chain, and investing in both charging infrastructure and battery-swapping ecosystems.

Providers are prioritizing customer experience, and financing options, creating a seamless transition to electric mobility. Integrating AI, IoT, and technology partnerships will elevate charging station efficiency, predictive analytics, and overall customer satisfaction, fostering long-term brand loyalty.

KEY GROWTH DRIVERS:

- Advancements in battery technology are revolutionizing energy storage, leading to increased adoption of battery-swapping stations.
- Rising fossil fuel costs and personal transportation preferences drive electric mobility demand. Robust charging infrastructure, financial incentives, and government support foster a conducive environment.
- Lithium-powered electric bikes prioritize efficiency, durability, and sustainability.

KEY PLAYERS:



KEY TRENDS:

The electric two-wheeler industry focuses on innovation in battery tech and braking systems, fostering brand loyalty through smartphone integration.



**Battery Technology
Advancements**



**Smart
Connectivity**



**Advanced
Braking System**



**Charging
Infrastructure**

Technology focus for 2-wheelers

OFFERINGS

Service Providers can prepare themselves with different initiatives and offerings in this area of technology targeted for 2-Wheeler to support different OEMs.

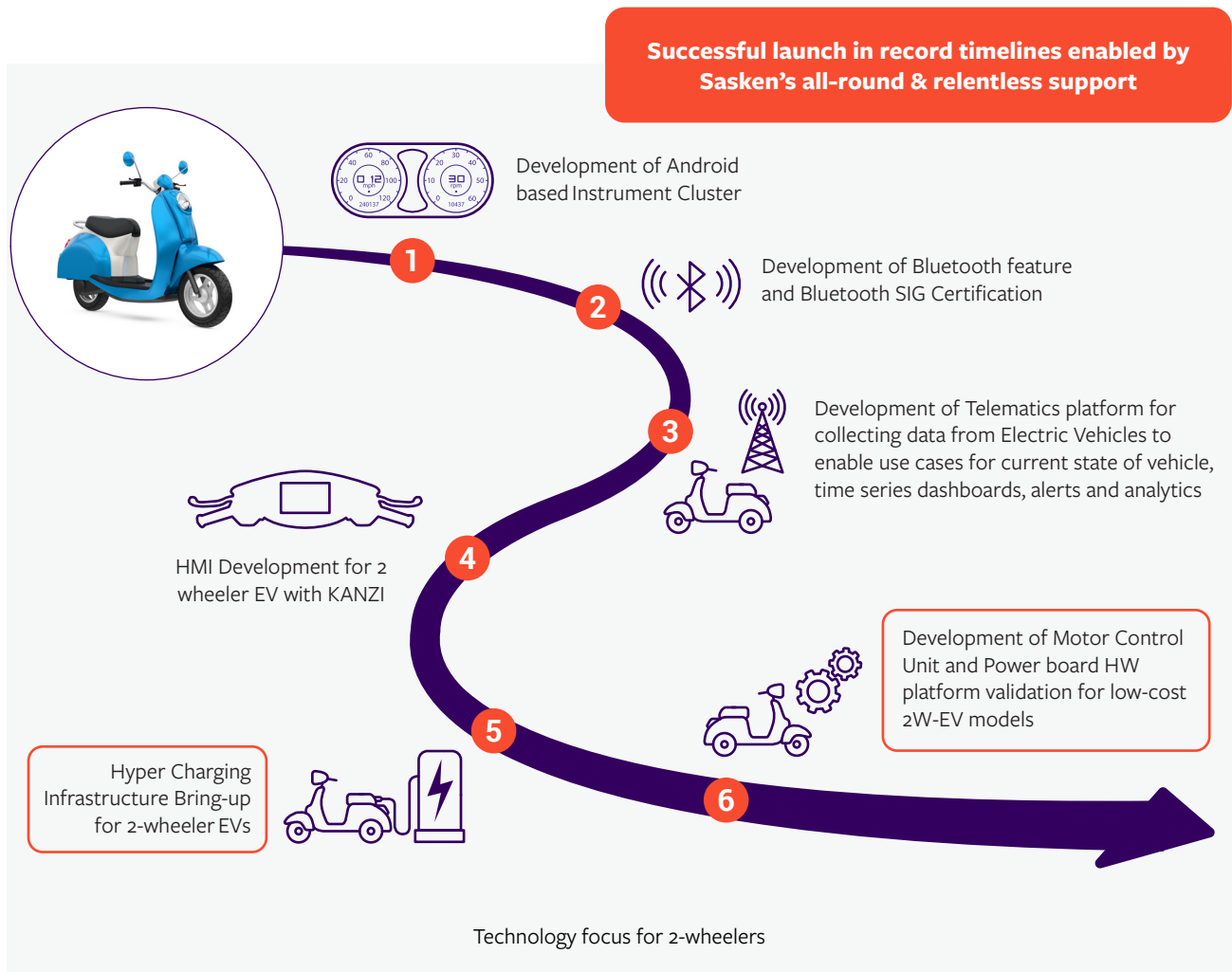
- Automotive offering related to ARAS with AIML and Battery Management System with digital solution for 2-Wheeler.
- Automotive offering related to CV2X/V2X targeted to 2-Wheeler.
- Digital and Automotive offering on eCall, Remote Services and Software update via OTA for 2-Wheeler.

CUSTOMERS/SUCCESS STORIES

SASKEN'S ENGAGEMENT WITH INDIAN 2-WHEELER OEM

Engaged with in development software for Digital Cluster, Body Control, Telematics, Companion App, Cloud applications, factory testing.

- **Body control ECU:**
 - Development of Motor Control Unit and Power board HW platform validation for low-cost 2W-EV models
- **ADAS:**
 - AVAS system Audio Integration
 - Haptic Warning Feedback – Lane Change, Over Speed, Possible Collision
- **Digital & Telematics:**
 - Desktop based diagnostic tool.
 - Implementing CI/CD pipeline for vehicle software
 - Companion app suite on Android and iOS
 - Charging infra support – Developed embedded charger controller and Cloud based application services.
 - Software Defined Vehicle - OTA feature development
 - Distributed Utility Apps
 - Vehicle Theft Warning



TELEMATICS SOLUTION DEVELOPMENT - EV MANUFACTURER -

Business Problem

Develop a telematics platform for collecting data from Electric Vehicles and Commercial Cars and enable use cases for current state of vehicle, time series dashboards, alerts, aggregation, and analytics.

Key Solution Highlights

Sasken is involved in:

- Architecture & Design of Telematics platform
- Development of use cases / features for different business units
- Manual and Automation testing

Tracking of electric vehicles (scooters) and associated telemetry data

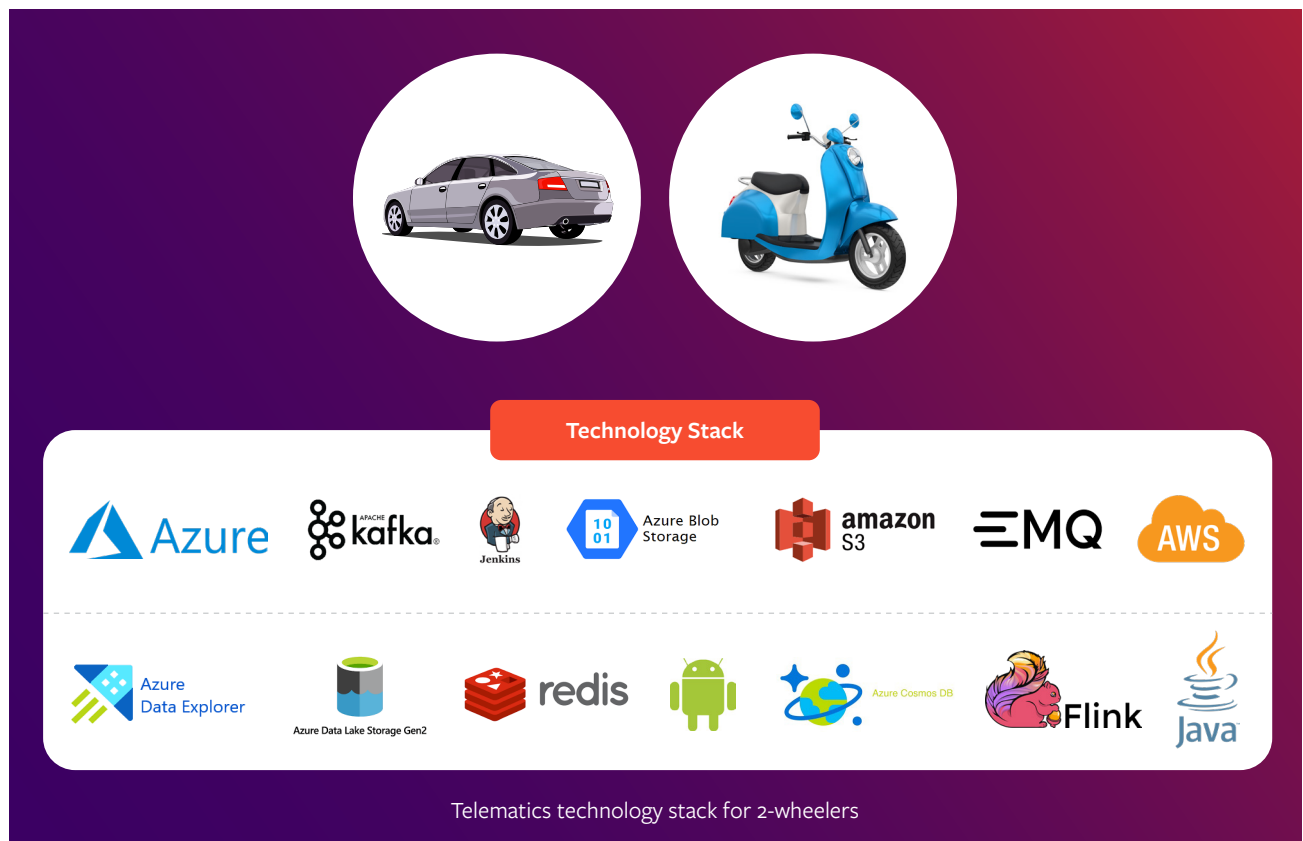
- Motor temperature
- Vehicle mode and speed
- Battery temperature
- GPS
- Errors and Warnings

Car companion app for consumers with feature subscriptions

- Vehicle health (service reminders, service booking)
- Vehicle security (live location, towing alerts, unauthorized access alerts, device tampering alerts, geofence violation alerts, time fence violation alerts)
- Vehicle assistance (insurance renewals, roadside assistance)

Outcome

Successfully completed the field trials and enabled customer to launch the Vehicles in the market.



MOTOR CONTROLLER UNIT FOR INDIAN 2-WHEELER OEM

Business Problem

Cost reduction + Local Sourcing for regulatory compliance + Product enhancement flexibility

Key Solution Highlights

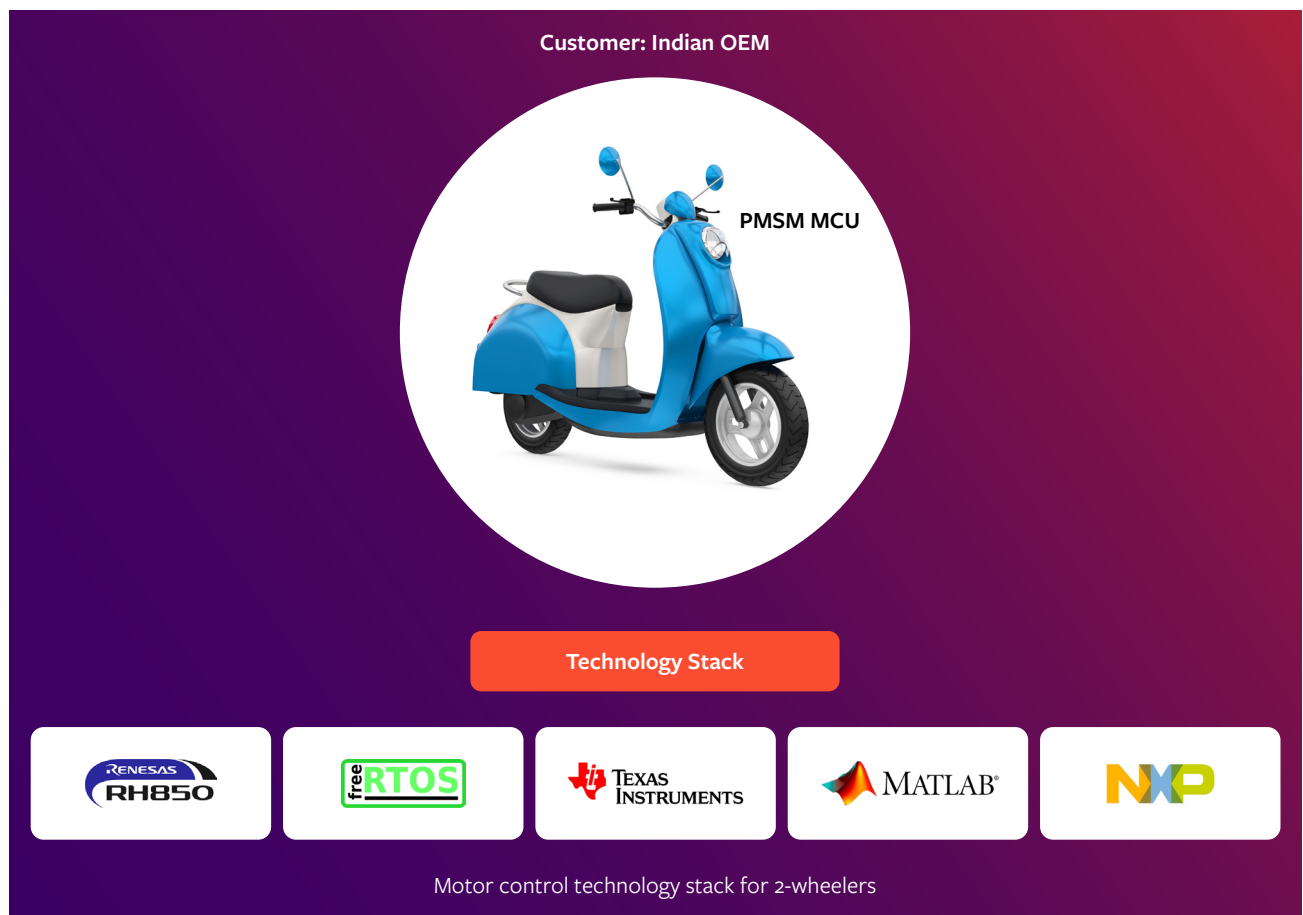
Co-engineered the solution:

- Requirement Scoping
- Feasibility analysis
- Cost benefit analysis
- MVP development
- Components selection
- Base SW development, FreeRTOS OS Porting, Power board HW platform validation, Pin muxing, Board bring-up, device driver validation.
- CAN & LIN stack porting
- Integration of core Matlab/Simulink model with Base SW
- Calibration tool development to fine tune motor control parameters
- Validation of MCU functions

Enhanced the solution with remote diagnostic support and SW update for faster customer issues resolution.

Outcome

Successful completion of MVP has led to a cost reduction of 50% of Motor Control System (MCU + Motor)



Conclusion

The whitepaper explores the transformative impact of technology on the future of the 2-wheeler industry, particularly focusing on Advanced Driver Assistance Systems (ADAS), Connectivity, and Electrification. It highlights the increasing significance of 2-wheelers in the automotive sector, especially in countries like India, owing to their affordability and efficiency in congested traffic. Despite their popularity, 2-wheelers face safety challenges, including accidents and fatalities, primarily due to reckless driving and inadequate safety gear usage. To address these issues, the paper proposes technological solutions such as Advanced Rider Assistance Systems (ARAS), Connectivity features, and Electrification. Additionally, it provides insights into the Indian electric two-wheeler market, showcasing momentum gained by electric vehicles and advancements in battery technology. The paper also outlines potential offerings for service providers and presents case studies illustrating successful engagements with 2-wheeler OEMs, ranging from telematics platform development to motor control unit enhancements. Lastly, it emphasizes the importance of capability building and suggests demonstration of proof of concepts (POCs) to showcase the efficacy of various technological solutions, offering a comprehensive roadmap for stakeholders to leverage emerging opportunities and tackle current challenges in the 2-wheeler industry.

About the Author

Mrinal Sarkar is the Senior System Architect at Sasken and leads the delivery of the company's leading automotive accounts. He also works closely with business teams on offerings and solutions. He has vast experience working as senior software architect for multiple automotive OEM programs across the globe. Apart from automotive, he also has widespread experience working in Telecomm and Consumer Electronics products.

About Sasken

At Sasken is at the forefront of Product Engineering and Digital Services, crafting solutions that redefine industries and shape the future of technology. Our deep engineering capabilities and technology patents have transformed Fortune 500 clients and powered billions of devices worldwide. We provide concept-to-market and chip-to-cognition R&D services to global leaders in the Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, Satcom, Telecom, and Transportation Industries.

Visit us at www.sasken.com



marketing@sasken.com | www.sasken.com

USA | Finland | Germany | Japan | India

Sasken technologies Ltd. All rights reserved.
Products and services mentioned herein are trademarks and services of Sasken Technologies Ltd., or the respective companies.

